MILITARY SPECIFICATION

CARTRIDGE, 40MM, HE, M384

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specifications covers the loading, assembling and packing for one type of cartridge designated as Cartridge, 40MM, HE, M384.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Method of
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of
MIL-P-60942	-	Primer Percussion for 40MM Ammunition

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-331	-	Fuze and Fuze Components, Environmental and Performance Test for
MIL-STD-1168	-	Lot Number of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

8886397	Cartridge, 40MM, HE, M384

PUBLICATIONS

ARMY

ET-8886397	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications. -The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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DEPARTMENT OF TRANSPORTATION REGULATIONS

49 CFR 171-178 - Department of Transportation Rules and Regulations for the Transportation of Explosives and other Dangerous Articles

(The Department of Transportation Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents. U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite, "49 CFR 171-178, (latest revision)".)

3. REQUIREMENTS

3.1 Material. -Materials shall be in accordance with the applicable drawings and specifications.

3.2 Cartridge. -The cartridge shall comply with all requirements specified on Drawing (Dwg) 8886397, all associated drawings and with all requirements specified in applicable specifications.

3.3 X-Ray examination of fuze assemblies prior to assembling to body assembly. -The fuze assembly shall be x-rayed for improperly assembled, missing parts, armed or partially armed (as evidenced by the detonator moving any degree from the safe position) when tested as specified in 4.4.7.

3.4 Transportation vibration. -The cartridge shall be safe and operable after test specified in 4.4.9.

3.5 Air pressure. -The cartridge shall withstand an air pressure of three (3) pounds per square inch (p.s.i.) for five (5) seconds, minimum (min) without a drop in pressure when tested as specified in 4.4.8.

3.6 Functioning.

3.6.1 Cartridge. -The cartridge shall function satisfactorily and the projectile shall have a mean velocity of 790 ± 15.0 feet per second (f.p.s.) and a standard deviation not exceeding 15.0 f.p.s. when tested as specified 4.4.10. The action time of the cartridge when tested as specified in 4.4.10 shall be 4.0 milliseconds, maximum.

3.6.2 Safety. -There shall be no premature burst (See 6.7) or evidence thereof in any of the test firings when tested as specified in 4.4.10.

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3.7 Workmanship. -All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.8 Submission of results of contractor-conducted examinations and tests.- The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract, the specification, or as requested by the Contracting Officer. (See 6.10)

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. -Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. -At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by code number, for all inspections performed,
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection. -Not applicable.

4.3 Inspection provisions.

4.3.1 Lot formation. -The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105.

Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of cartridges shall contain:

a. Cartridge case assembly metal parts from one lot interfix number from one manufacturer.
b. Metal parts from one lot interfix number from one manufacturer.
c. Fuzes from no more than two (2) consecutively produced serial lots from one manufacturer.
d. Primers from one lot number from one manufacturer.
e. Propellant from not more than one (1) lot number from one manufacturer.
f. Composition A5 from one lot interfix number from one manufacturer.

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g. Cartridges in less than full belt quantities from any acceptable lot of cartridges shall be held until a cumulative amount of 3000 cartridges are obtained or until 1 year has elapsed. These cartridges shall be linked, packed and shipped in multiples of 50 as a conglomerate lot. No further ballistic firing need be performed on these cartridges, since these cartridges came from acceptable lots. Acceptable lots are not accepted on waiver. If at the end of 1 year, 3000 cartridges are not obtained, the cumulative amount of cartridges shall be linked, packed and shipped in multiples of 50 and the remaining cartridges shall be destroyed.

4.3.2 Examination. -Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105), except that, inspection for Critical defects shall be 100 percent. Contractor's sampling plans, if used, shall be approved by the Government, and shall provide, as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect except where 100% inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cartridge case assembly, prior to loading (See Dwg 8886327 covering a detail of Dwg 8886397). -

Categories	Defects	Method of Inspection	Code No (See 6.2)
Critical:	None defined		
Major	100% Inspection		
101.	Closing cup missing	Visual	06001
Minor	AQL 0.65 Percent		
201.	Talc assembly missing	Visual	01001

4.3.2.2 Cartridge case assembly, prior to inserting base plug and primer (See Dwg 8886327 covering a detail of Dwg 8886397). -

Categories	Defects	Method of Inspection	Code No
Critical			
1.	Spilled powder from cartridge case	Visual	02001
2.	Propellant charge weight obviously incorrect (See 6.9)	Visual	02002

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Categories	Defects	Method of Inspection	Code No
Major	100% Inspection		
101.	Propellant charge weight obviously incorrect (See 6.9)	Visual	02003

Minor: None defined

4.3.2.3 Cartridge case assembly (See Dwg 8896327 covering a detail of Dwg 8886397). -

Categories	Defects	Method of Inspection	Code No
Critical			
1.	Primer above flush with base plug	Gage	03001
Major	AQL 1.00 Percent		
101.	Base plug above flush	Gage	03002
102.	Depth to base plug, max	Gage	03003
103.	Depth to primer, max from base plug	Gage	03004
104.	Primer damaged	Visual	03006
105.	Excessive sealing compound on exterior surface of primer	Visual	03007
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.7)	Visual	03008

4.3.2.4 Body loading assembly (See Dwg 8886400 covering a detail of Dwg 8886398). -

Categories	Defects	Method of Inspection	Code No
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Depth to charge	Gage	04001
102.	Diameter of rotating bands	Gage	04002
Minor	AQL 1.50 Percent		
201.	Threads damaged	Visual	04003
202.	Evidence of poor workmanship (See 3.7)	Visual	04004
203.	Charge parallel to rim of body	Gage	04005

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4.3.2.5 Projectile, prior to crimping cartridge case assembly (See Dwg 8886397). -

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor	AQL 0.65 Percent		
201.	O-ring (packing) missing in groove	Visual	05001

4.3.2.6 Cartridge (See Dwg. 8886397). -

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Gap, min	Gage	06001
102.	O-ring exposed	Visual	06002
103.	Gap, rear of rotating bands	Gage	06003
Minor	AQL 1.50 Percent		
201.	Excessive adhesive on exterior surfaces	Visual	06004
202.	Marking missing	Visual	06005
203.	Evidence of poor workmanship (See 3.7)	Visual	06006

4.3.2.7 Linked ammunition prior to packing (See Dwg 8886405). -

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Link improper (Anyone of the six (6) tabs of link not in groove; link inverted; link damaged; link distorted)	Visual	07003
Minor:	None defined		

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4.3.2.8 Unsealed fiber box (See Dwg 9251996). -

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Cartridge inverted	Visual	08001
102.	Incorrect number of cartridges	Visual	08002
103.	Fillers missing	Visual	08003
Minor:	None defined		

4.3.2.9 Sealed fiber box, prior to sealing bag (See Dwg 9251996). -

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Container damaged	Visual	09001
102.	Corners not blunted	Visual	09002
Minor	AQL 1.00 Percent		
201.	Contents loose	Manual	09003
202.	Label missing or unidentifiable	Visual	09004

4.3.2.10 Unsealed wooden packing box (See Dwg 9251995). -

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Marking of bag misleading or unidentifiable	Visual	10001
102.	Bag torn or perforated	Visual	10002
103.	Bag improperly sealed	Visual	10003
Minor:	None defined		

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4.3.2.11 Sealed wooden packing box (See Dwg 9251995). -

Categories	Defects	Method of Inspection	Code No,
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Box damaged	Visual	11001
Minor	AQL 1.50 Percent		
201.	Contents loose	Manual	11002
202.	Car seal missing or improperly positioned	Visual	11003
203.	Marking misleading or unidentifiable	Visual	11004

4.3.3 Testing.

4.3.3.1 Check test for deterioration of primers (See MIL-P-60942) Major Defect, Code No. 12001. -If the total time between original acceptance of any lot and the assembly of that lot into the cartridge exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-P-60942 immediately before the primer lot is assembled into the cartridge. This test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the primers into the cartridge. (See 6.4)

4.3.3.2 Propellant weight of case, cartridge (See Dwg 8886327) Major Defect, Code No. 13001. -The propellant weight shall be determined and then check weighed 100 percent. The check weighing shall be accomplished independently of the original weighing or determination, using a different balance from that used to make the original weighing and if performed manually, shall be performed by another operator. Any charge which fails to comply with the requirement specified on the applicable drawing shall be classified defective and removed from the lot.

4.3.3.3 Security of projectile (See Dwg 8886397) Major Defect, Code No. 14001. -The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter J, single plan, with an AQL of 0.65 percent. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

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4.3.3.4 Security of crimp of base plug on cartridge case assembly (See Dwg. 8886327) and action time of cartridge case assembly (See Para. 3.6.1), Major Defect Code No. 15001. -Five (5) base plug and primer assemblies shall be randomly selected every four (4) hours from each manual primer assembly machine or twenty (20) base plug and primer assemblies shall be randomly selected every four (4) hours from each automatic primer assembly. These assemblies shall be assembled to cartridge cases for test of "Security of crimp of base plug and action time" determination. The base plug and primer assemblies produced by each primer assembly machine shall be kept segregated and identified. They shall not be used in production until the above tests have been performed. If any base plug moves more than .005 above flush with rear of case, or the action time of any cartridge case assembly exceeds 4.0 milliseconds, the four (4) hours production from each manual primer assembly machine or each automatic primer assembly machine involved shall be rejected. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.5 Specific gravity of each increment of body loading assembly (See Dwg 8886400), Major defect, Code No. 16001. -Two (2) sample cup assemblies of each increment, shall be selected from each four (4) hours production and subjected to this test. If any sample fails to comply with specific gravity requirement specified on the applicable drawing, the four (4) hours production represented by the sample shall be rejected. The test shall be performed as specified in 4.4.3.

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4.3.3.6 Security of fuze on projectile assembly (See Dwg 8886398) Major Defect, Code No. 17001. -The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter K, single plan, with an AQL of 0.40 percent. Samples shall be randomly selected after assembly and the test performed before adhesive hardens. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.7 Total length of cartridge (See Dwg 8886397) Major Defect, Code No. 18001. -Each cartridge shall be subjected to this test. If any cartridge fails to comply with the requirement specified on the applicable drawing, it shall be classified defective and removed from the lot. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

4.3.3.8 Chamber gage failure (See Dwg 8886397) Major Defect, Code No. 19001. -Each cartridge shall be subjected to this test. If any cartridge fails to comply with the requirement specified on the applicable drawing, it shall be classified defective and removed from the lot. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4.

4.3.3.9 X-ray examination of fuze assemblies prior to assembling to body assembly (See 3.3), Code No. 25001. -Each fuze assembly prior to assembling to body assembly shall be subjected to this test. If any fuze assembly shows evidence of being improperly assembled, having missing parts, being armed or partially armed, it shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.7.

4.3.3.10 Air pressure (See Para 3.5) Major Defect, Code No. 20001. -The sampling plan for this test shall be in accordance with MIL-STD-105, Code Letter P, single plan, with an AQL of 0.015 percent. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.11 Transportation vibration of cartridge prior to subsequent loading (See paragraph 3.4), Major defect, Code No. 26001. -Three-hundred (300) fuze assemblies shall be selected from each fuze lot, x-rayed, assembled as M384 cartridges, pack in their regular shipping containers and subjected to this test. The test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4.

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4.3.3.12 Functioning, prior to subsequent loading (See Paragraph 3.6), Major Defect, Code No. 27001. -After completion of test specified in 4.3.3.11, the three-hundred (300) cartridges shall be removed from their regular shipping containers and subjected to this test. The fuze lot shall be rejected if twelve (12) or more fail high order detonation at initial impact. The test shall be performed as specified in 4.4.10 except the samples shall not be shipped to a Government Proving Ground.

4.3.3.13 Functioning (See Para 3.6) Major Defect, Code No. 21001. - Beginning with the first lot produced and continuing until three (3) consecutive lots have been accepted under the criteria specified, the Government inspector shall select three-hundred (300) cartridges from each lot for this test. The samples shall be selected at random prior to linking and then linked.

4.3.3.13.1 Rejection. -The lot shall be rejected if:

a. Thirteen (13) or more fail high order detonation at initial impact.
b. Any primer blows back (as evidenced by perforation of the primer cup or smoke deposit on base plug or cartridge case base).
c. Any premature burst occurs (See 6.7).
d. Two (2) or more primers misfire.
e. Any projectile sticks in gun bore.
f. Mean velocity and/or standard deviation fail to meet the applicable requirement.
g. Three (3) or more base plugs are more than .005 above flush with rear of cartridge case (See 4.4.11) or one (1) or more base plugs exceeds .010 with rear of cartridge case or jams the automatic weapon.
h. The action time of three (3) or more rounds are over four (4) milliseconds or if one (1) round is over ten (10) milliseconds.

If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved launcher. If there is significant difference in the standard deviation and mean velocity of the control lot from that previously obtained for control rounds, then a new approved launcher shall be obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on retest. If there is no significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

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4.3.3.13.1.1 If ten (10) or twelve (12) assemblies fail high order detonation initial impact, a second sample of 300 cartridges shall be forwarded and tested. If the combined number of rejects in the first and second sample is twenty-one (21) or more, the lot shall be rejected.

4.3.3.13.2 Suitability test plan (Informational test only) Code No. 21002. -Beginning with the first lot produced by each producer and continuing until that producer has submitted three (3) consecutive lots that are accepted in accordance with 4.3.3.13, an extra sample of ninety (90) rounds shall be selected from each lot and pre-conditioned and tested as follows:

a. The ninety (90) rounds shall be packaged in their regular shipping containers and subjected to transportation vibration environment in accordance with MIL-STD-331, Test No. 104.

b. After transportation vibration, the rounds shall be unpacked and subjected to JAN temperature and humidity cycle (14 days) specified in MIL-STD-331, Test No. 105.

c. After temperature and humidity cycle, the rounds shall be temperature conditioned and fired as follows:

30 rounds +165 degrees F. ± 5 degrees F.
30 rounds ambient (between +40 and +110 degrees F.)
30 rounds -65 degrees F. + 5 degrees F.

d. Fire the rounds in accordance with paragraph 4.4.10 and with TECOM Material Test Procedures, TECP 700-700 Volume 4. Test results, failure analysis reports, and other information as requested, shall be forwarded to Picatinny Arsenal, ATTN: SMUPA-ND3 for evaluation.

4.3.3.14 Functioning (See Para 3.6) Major Defect, Regular Production, Code No. 22001. -After three (3) consecutive lots have met the criteria of 4.3.3.13, the Government inspector shall select one-hundred (100) cartridges from each lot for this test. If two (2) or more lots fail the criteria of 4.3.3,14.1, the consecutive lots thereafter shall revert to the criteria of paragraph 4.3.3.13.

4.3.3.14.1 Rejection. -The lot shall be rejected if:

a. Seven (7) or more fail high order detonation at initial impact.

b. Any primer blows back (as evidenced by perforation of the primer cup or smoke deposit on base plug or cartridge case base).

c. Any premature burst occurs (See 6.7).

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d. Two (2) or more primers misfire.
e. Any projectile sticks in gun bore.
f. Mean velocity and/or standard deviation fail to meet the applicable requirement.
g. Three (3) or more base plugs are more than .005 above flush with rear of cartridge case (See 4.4.11) or one (1) or more base plugs exceeds .010 with rear of cartridge case or jams the automatic weapon.
h. The action time of one (1) or two (2) rounds is over four (4) milliseconds, with no round exceeding ten (10) milliseconds, an additional 200 rounds shall be tested. The results for the 300 rounds shall be combined. If three (3) or more rounds are four (4) milliseconds or if one (1) round is over ten (10) milliseconds, the lot shall be rejected.

4.3.3.14.1.1 If four (4) to six (6) assemblies fail high order detonation at initial impact, a second sample of one hundred (100) cartridges shall be forwarded and tested. If the combined number of rejects in the first and second sample is nine (9) or more, the lot shall be rejected.

If the launcher is suspected of being the cause of rejection for the mean velocity and standard deviation phase, sufficient cartridges from a controlled lot shall be fired from the same approved launcher. If there is significant difference in the standard deviation and mean velocity of the control lot from that previously obtained for control rounds, then a new approved launcher shall be obtained and the lot of ammunition retested. If all cartridges function properly on original test, then only the velocity will be considered on retest. If there is no significant difference in the standard deviation or mean velocity of the control lot, then the lot shall be rejected.

4.3.3.15 Heat seal test of packing seals (See Dwg 9251996) Major Defect, Code No. 23001. -The samples for this test shall be selected in accordance with MIL-P-116. Rejection shall be in accordance with MIL-P-116. The test shall be performed as specified in 4.4.12.

4.3.3.16 Rapid fire functioning, Major Defect, Code No. (24001). -One hundred (100) cartridges shall be randomly selected from a completed lot and subjected to this test. If any gun stoppage occurs or if any projectile sticks in the gun bore, the lot shall be rejected. The test shall be performed as specified in 4.4.13 using equipment in accordance with 4.3.4.

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4.3.4 Inspection equipment. -Equipment Tabulation (ET) number 8886397 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation. -Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Security of crimp of base plug on cartridge case assembly. -The cartridge case assembly shall be assembled to a projectile and fired. Measurement for base plug more than .005 above flush from the rear of the cartridge case shall be measured from the highest point on the base plug to a point on cartridge case directly beside it, radially. Observation shall be made for action time using equipment specified on the Equipment List.

4.4.2 Security of projectile. -The cartridge shall be placed in the fixture shown on the Equipment List and the axial tension specified on Dwg 8886397 shall be applied. Cartridge case assemblies shall not be reused. Projectile assemblies may be reused. The cartridge shall be pulled until separation occurs and the data recorded.

4.4.3 Specific gravity of the charge of body loading assembly. -This test shall be conducted by any method approved by the Contracting Officer.

4.4.4 Security of fuze on projectile assembly. -The projectile assembly shall be placed in the fixture shown on the Equipment List and the minimum torque shall be applied. Observation shall be made for the requirement specified on Dwg 8886398. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.6 Total length of cartridge. -Using the equipment specified on the Equipment List, the cartridge must meet the requirement specified on Dwg 8886397. This test is a non-destructive test. Parts so tested may be returned to the lot.

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4.4.6 Chamber gage failure. -The cartridge shall be placed in the chamber shown on the Equipment List and must meet the requirement specified on Dwg 8886397. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.7 X-ray examination of fuze assembly prior to assembling to body assembly. -Improperly assembled, missing parts, armed or partially armed fuze assemblies shall be determined by X-ray equipment or any other method satisfactory to the Contracting Officer. This test shall be conducted in two (2) planes. From the top of the fuze assembly down and from the side.

4.4.8 Air pressure. -The cartridge shall be placed in the fixture shown on the Equipment List and a measured quantity of air shall be applied to produce the required air pressure. This test is a non-destructive test. Parts so tested may be returned to the lot.

4.4.9 Transportation vibration of cartridge, prior to subsequent loading. - The cartridge shall be tested as specified in Method 104,Procedure II of MIL-STD-331. Cartridges shall be subjected to subsequent testing.

4.4.10 Functioning. -This test shall be conducted at a Government owned proving ground. The rounds being tested, including the standard control rounds, shall be temperature conditioned at 70°F ± 5°F for at least eight (8) hours, then fired within fifteen (15) minutes after being removed from temperature soak. The test and standard control rounds shall be fired at a rate not to exceed two rounds per minute using an XM129 launcher as specified on the Equipment List. The XM129 launcher shall have been "broken-in" if new, with at least forty (40) rounds and placed on an approved mount for running the test. At least 6 gun-warmers shall be rapid fired at the start of the test. After the gun-warmers have been fired, twelve (12) of the standard control rounds shall be fired. These shall be followed by the test samples and then twelve (12) more of the standard control rounds. Distance to impact 200 meters plus or minus (±) 5 meters, impact media, coarse sand, ten (10) inches minimum thickness at point of impact. The angle of firing shall be adjusted to give an impact of approximately 25 degrees. Rake sand as required such that individual impact craters do not overlap Sand will be raked after each dud. If a round falls outside the specified impact area it shall be declared a "no-test" round. The velocity shall be measured at 30 feet from the muzzle of the launcher and check measured at the same base line. The monitoring devices and time recorders shall be a dual system set-up for recording two (2) separate readings and shall be within 3 feet per second. If the two (2) readings differ more than 3 feet per second (individual shots) the results shall be discarded and another round fired in its place. When proving ground reports results for functioning, the number of all "no-test" rounds shall also be reported. "No-test" rounds shall be reported as "no-test" rounds outside the impact-area or "no-test" rounds for velocity readings. Observations shall be

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made for functioning and results calculated and recorded for (a) cartridge action time, using the equipment specified on the Equipment List and (b) mean velocity and standard deviation (See 6.6). The mean velocity of the lot being tested shall be corrected by the correction factor determined for the particular test launcher being used on the day of the test firing. When velocity values are suspected to be outliers, they shall be tested using the statistical procedure approved by the Government proving ground and the mean velocity and standard deviation adjusted accordingly. However, no more than two (2) on the high side and no more than (2) on the low side shall be considered for exclusion. The correction factor for the test launcher shall be determined by subtracting the mean velocity of the standard control rounds fired during the test from the standard velocity previously established for the standard control rounds. Either a positive or a negative correction factor will result depending on the mean velocity of the standard control rounds fired during the test. Algebraically add the resultant correction factor to the mean velocity of the test rounds.

4.4.10.1 Test Validity. -If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.4.11 Movement of base plug. -The movement of the base plug shall be measured from the highest point on the base plug to a point on cartridge case directly beside it, radially.

4.4.12 Heat seal test of packing seals. -This test shall be conducted accordance with MIL-P-116. Seals so tested shall not be returned to the lot.

4.4.13 Rapid fire functioning. -This test shall be conducted at a Government owned proving ground. The cartridge shall be fired from the XM129 launcher specified on the Equipment List in burst of approximately ten (10) rounds. Observation shall be made for gun stoppage and projectile stuck in the gun bore.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. -The cartridge shall be preserved and packaged in accordance with Dwg 9251996.

5.2 Packing.

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5.2.1 Level A. -The cartridge preserved and packaged in accordance with 5.1.1, shall be packed in accordance with Dwg 9251995.

5.3 Marking. -Marking shall be in accordance with Dwg 8796522 and Department of Transportation Code of Federal Regulations 49 CFR 171-178.

5.4 Shipping. -When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data. -Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provision for check testing of primer (See 6.4)

6.2 Inspection code numbers. -The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment. -The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists. -Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These Contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes. -The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

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6.3.3 Army designs. -Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contract designs. -Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design. -All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally within one month after receipt.

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6.4 Cost of check test. - The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The test shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies or to the contractor assembling the primer assemblies into the cartridge, and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc, permitted while the primer assembly lot was under the jurisdication of either contractor.

6.5 Submission of test data. -In addition to the normal distribution of records, when the cartridge is procured by the U.S. Army Munitions Command, one (1) copy of all ballistic data and ammunition data cards shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.6 Standard deviation shall be calculated from the following formula:

$$S = \sqrt{\frac{\Sigma(X - \bar{X})^2}{n - 1}}$$

where:

X = Each individual value
$\bar{X}$ = Sample arithmetic mean $\frac{\Sigma X}{n}$
n = Sample size
$X - \bar{X})^2$ = The sum of the squares of the differences between the sample mean and each individual values.
n - 1 = One less than the sample

6.6.1 For a faster and easier method of arriving at the standard deviation, the following alternate method may be used:

Other approved standard deviation formulas may be used.

6.7 Premature burst. -A premature burst is considered to have occurred if the round functions in gun bore or in flight.

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6.8 Combining of proving ground tests. -When the contractor for the cartridge is also the contractor for one or more of the component thereof, the proving ground test of the contractor may be combined with the proving ground test of the cartridge, to save. expense, upon agreement between the procuring activity and the contractor. In cases where the cartridge specification does not cover all the proving round tests specified for the component, the additional tests specified in the component specifications shall be conducted.

6.9 If any cartridge case, prior to insertion of the base plug and primer, is suspected of containing an underweight or overweight propellant charge, based on visual inspection, it will be removed from the lot and weighed on suitable scales. Any cartridge case found with less than 75 percent or more than 125 percent of the assessed propellant load will be classified as a Critical defect and removed from the lot. Any cartridge case found to have a propellant weight out of drawing tolerance but within ± 25 percent of the assessed propellant load will be classified as a major defect and removed from the lot.

6.10 Submission of results of contractor-conducted examinations and tests. - Unless otherwise specified by the Contractor Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1310-A-116

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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PICATINNY ARSENAL
DOVER, NEW JERSEY 07801

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